

# Examination Java Programming

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Consider the following classes implementing ephemeral stacks:

```
public class Stack<Key extends Comparable<Key>> {  
    protected Cell<Key> top;  
    public Stack () { top = null; }  
}
```

```
public class Cell<Key extends Comparable<Key>> {  
    protected Key value;  
    protected Cell<Key> next;  
    public Cell (final Key key, final Cell<Key> cell) {  
        value = key; next = cell; }  
}
```

Extend the class `Stack<Key>` with two methods defined as follows.

1. The method `rm_fst` (*remove first*) whose signature is

```
public void rm_fst (final Key k);
```

performs a linear search in `this`: if the key `k` is absent, nothing is done; otherwise, the (first) cell containing `k` is taken off `this`.

2. The method `rm_lst` (*remove last*) whose signature is

```
public void rm_lst (final Key k);
```

removes the last cell containing `k`, and, otherwise, do nothing.

The main class for testing is

```
public class EphObj {
    public static void main (String[] args) {
        Stack<Integer> s = new Stack<Integer>();
        s.push(5); s.push(1); s.push(3); s.push(1);
        s.push(0); s.push(3);
        s.print(); // 3 0 1 3 1 5
        s.rm_fst(3);
        s.print(); // 0 1 3 1 5
        s.rm_fst(1);
        s.print(); // 0 3 1 5
        s.rm_fst(5);
        s.print(); // 0 3 1
        s.rm_fst(6);
        s.print(); // 0 3 1

        s.push(1); s.push(5); s.push(3);
        s.print(); // 3 5 1 0 3 1
        s.rm_lst(3);
        s.print(); // 3 5 1 0 1
        s.rm_lst(1);
        s.print(); // 3 5 1 0
        s.rm_lst(3);
        s.print(); // 5 1 0
        s.rm_lst(4);
        s.print(); // 5 1 0
    }
}
```